SLRMA outlines

Telemedicine Technology in Breast Cancer Management: A Systematic Review of Clinical Applications, Effectiveness, and Limitations

1. Abstract

- Background
- Objective
- Methods
- Results
- Conclusion
- Keywords

2. Introduction

2.1. Background and global burden of breast cancer

2.2. Overview of breast cancer management and follow-up care

2.3. Evolution of telemedicine and digital health in oncology

2.4. Clinical applications of telemedicine in breast cancer management

2.5. Rationale for the systematic review

2.6. Research gap and significance of the review

2.7. Aim and objectives

3. Materials and Methods

3.1. Study Design and Reporting Guidelines

- Systematic review design
- PRISMA 2020 reporting guidelines
- Protocol development and registration, if applicable

3.2. Research Questions

- Types of telemedicine technologies used in breast cancer management
- Effectiveness of telemedicine in improving clinical and patient-reported outcomes
- Limitations and barriers to implementation

3.3. Information Sources and Databases

- PubMed
- Scopus
- Search date and coverage period

3.4. Search Strategy

- Development of database-specific search strings
- Keywords and Boolean operators
- Controlled vocabulary, including MeSH terms in PubMed
- Supplementary searching of reference lists, if undertaken

3.5. Eligibility Criteria

- Inclusion criteria
- Exclusion criteria
- Population, intervention, comparator, outcomes, and study design

3.6. Study Selection Process

- Identification of records
- Duplicate removal
- Title and abstract screening
- Full-text screening
- Documentation of exclusion reasons

3.7. Data Extraction

- Study characteristics
- Population characteristics
- Telemedicine technology and clinical application
- Comparator and outcome measures
- Main findings and limitations

3.8. Risk-of-Bias and Quality Assessment

- RoB 2 for randomized controlled trials
- ROBINS-I for non-randomized intervention studies
- Appropriate appraisal tools for observational and qualitative studies

3.9. Data Synthesis and Analysis

- Narrative synthesis
- Thematic grouping of clinical applications
- Quantitative synthesis, if appropriate
- Statistical methods for meta-analysis, if feasible

4. Results

4.1. Study Selection

- Identification, screening, eligibility, and final inclusion
- PRISMA flow diagram

4.2. Characteristics of Included Studies

- Publication year and country
- Study design and setting
- Sample size and patient characteristics
- Cancer stage and treatment status

4.3. Types of Telemedicine Technologies

- Video consultations and telephone follow-ups
- Mobile health applications
- Web-based platforms and secure messaging
- Remote patient monitoring
- Electronic patient-reported outcome systems
- Hybrid telemedicine models

4.4. Clinical Applications in Breast Cancer Management

- Treatment-related consultations and monitoring
- Chemotherapy-related symptom management
- Postoperative follow-up and recovery
- Survivorship care
- Psychological support and counselling
- Palliative and supportive care

- Patient education and self-management

4.5. Effectiveness of Telemedicine

- Clinical outcomes and symptom control
- Treatment adherence and continuity of care
- Quality of life and psychological well-being
- Patient satisfaction and acceptability
- Healthcare utilization and unplanned hospital visits
- Access to care and reduced travel burden
- Safety and timely escalation to in-person assessment

4.6. Limitations and Implementation Barriers

- Digital literacy and technological accessibility
- Internet connectivity and infrastructure
- Privacy and data security
- Limitations of remote physical examination
- Socioeconomic and geographical inequalities
- Communication challenges
- Cost, reimbursement, and healthcare workforce constraints

4.7. Risk-of-Bias Assessment

- Overall methodological quality
- Domain-level risk-of-bias findings
- Potential implications for confidence in the evidence

4.8. Summary of Key Findings

- Consistent findings across studies
- Conflicting or inconclusive evidence
- Areas with insufficient evidence

5. Discussion

5.1. Principal Findings

5.2. Interpretation of Telemedicine Effectiveness in Breast Cancer Care

5.3. Comparison of Different Telemedicine Modalities

5.4. Clinical Implications for Breast Cancer Management

5.5. Patient-Centred Care and Healthcare Accessibility

5.6. Safety, Equity, Privacy, and Implementation Challenges

5.7. Research Gaps and Future Directions

5.8. Strengths and Limitations of the Review

6. Conclusion

- Summary of the main findings
- Implications for clinical practice
- Recommendations for future research and implementation

7. Declarations

- Funding
- Conflicts of interest
- Data availability statement
- Author contributions
- Ethics statement, where applicable

8. References

- References formatted according to the target journal's requirements.